

MAHAN BALOOEI

Modena, Italy · mahan.balooei@studio.unibo.it · +39 351 748 8239 · LinkedIn · GitHub

PROFILE

MSc Bioinformatics student at the University of Bologna with a background in Medical Laboratory Sciences and a primary focus on protein sequence modelling, protein language models, and ML-based functional annotation. Experienced with ESM-2 embeddings, signal peptide prediction, HMMER/MMseqs2 workflows, DNA methylation analysis in R/Bioconductor, and reproducible ML pipelines. Broader interests include AI applications in proteomics, transcriptomics, single-cell omics, and biomedical data science. Seeking a research or MSc thesis internship, with flexibility regarding the start date.

EDUCATION

University of Bologna — MSc in Bioinformatics Oct 2024 – Present

Coursework: sequence alignment, RNA-seq, microarrays, statistical analysis in R; ML/DL with scikit-learn & TensorFlow; SQL/graph queries, Docker, HPC/Slurm.

Scientific Focus: protein language models, protein sequence representation learning, protein functional annotation, sequence-function prediction, and remote homology detection. Interested in extending this toward transcriptomics and single-cell omics.

Iran University of Medical Sciences — BSc in Medical Laboratory Sciences Sept 2019 – Jul 2023

Relevant Coursework: Molecular Diagnostics, Immunoassays, Clinical Chemistry, Haematology, Microbiology, Virology.

PUBLICATIONS

Mechanism-based targeting of lactate dehydrogenase tetramerization by naturally derived compounds: A survey of molecular docking, molecular dynamics, and QSAR analysis Sep 2025

Computers in Biology and Medicine, 2025

- Investigated plant-derived compounds as LDH inhibitors for anticancer applications; identified 5 top candidates from pomegranate and sumac with docking scores >8 kcal/mol and stable MD simulations over 100 ns.
- Applied molecular docking, dynamics simulations, and QSAR modelling.
- Published in (Elsevier, 2025). [Article link](#)

RESEARCH PROJECTS & COMPUTATIONAL EXPERIENCE

Protein Language Model-Based Signal Peptide Prediction Sep 2025 – Nov 2025

MSc Bioinformatics Project - independently implemented

- Built an end-to-end protein sequence modelling pipeline for eukaryotic signal-peptide prediction using high-confidence UniProtKB/Swiss-Prot annotations.
- Applied MMseqs2 clustering for redundancy reduction and constructed biologically meaningful train/benchmark splits.
- Compared von Heijne PSWM, biochemical-feature SVM, CNN-BiLSTM, and ESM-2 embedding-based models.
- Achieved $F1 = 0.962$ and $MCC = 0.957$ on a held-out benchmark set using the ESM-2-based model.
- Analyzed amino-acid composition, signal-peptide lengths, kingdom-specific bias, and cleavage-site motifs for biological interpretation.
- [GitHub Repository](#)

Protein Domain and Functional Annotation Using HMMER and MMseqs2

Jun 2025 – Jul 2025

- Built an automated protein domain annotation workflow using profile HMMs, HMMER, and MMseqs2 for large-scale sequence classification.
- Processed protein structure-derived alignments to support domain-level annotation and sequence-based functional inference.
- Compared profile-based and sequence-similarity-based classification workflows, achieving MCC = 0.997 and MCC = 0.991, respectively.
- Connected classical homology and profile-HMM approaches to modern protein representation learning tasks such as remote homology detection and protein family classification.
- [GitHub Repository](#)

Computational Epigenomics Analysis of DNA Methylation (Illumina 450K)

Jun 2025 – Jul 2025

- Conducted an R/Bioconductor workflow for Illumina 450K DNA methylation array analysis.
- Performed quality control, normalization, PCA-based batch-effect assessment, and differential methylation analysis.
- Identified Differentially Methylated Regions (DMRs) and visualized methylation patterns using heatmaps, volcano plots, and exploratory statistical summaries.
- [GitHub Repository](#)

Chest X-Ray Pneumonia Detection via Transfer Learning

Mar 2026 – Apr 2026

- Compared ResNet50, EfficientNetB0, and DenseNet-121 XRV for pneumonia classification.
- Addressed class imbalance and evaluated threshold-selection strategies using clinically relevant metrics.
- Applied ensembling, test-time augmentation, and Grad-CAM-based interpretability.
- DenseNet-121 XRV achieved the strongest performance under the maximum-MCC threshold strategy.
- [GitHub Repository](#)

SKILLS

Functional Genomics & Epigenomics: DNA methylation analysis (Illumina 450K arrays, minfi), differential methylation region (DMR) detection, PCA-based batch-effect assessment, R/Bioconductor.

Protein Bioinformatics & Sequence Analysis: ESM-2 protein language model embeddings, protein representation learning, signal peptide prediction, sequence-function prediction, protein-domain annotation, HMMER, MMseqs2, BLAST, Clustal, UniProt, PDB.

Machine Learning & Deep Learning: PyTorch, scikit-learn, TensorFlow, SVMs, CNN-BiLSTM, transfer learning, representation learning, model evaluation, threshold optimization, class-imbalance handling, ensembling, and Grad-CAM-based interpretability.

Reproducible Workflows & Tools: Python, R, Bash/Linux, SQL, Git/GitHub, Docker, HPC/Slurm, R Markdown, Jupyter Notebooks, conda environments.

WORK EXPERIENCE

Clinical Laboratory Internships

Tehran, 2022–2023

- Rotations in haematology, microbiology, clinical chemistry, and biochemistry across hospital and diagnostic laboratory settings.

LANGUAGES

English: Proficient (TOEFL iBT 93, CEFR B2)

Italian: Basic communication

Persian (Farsi): Native